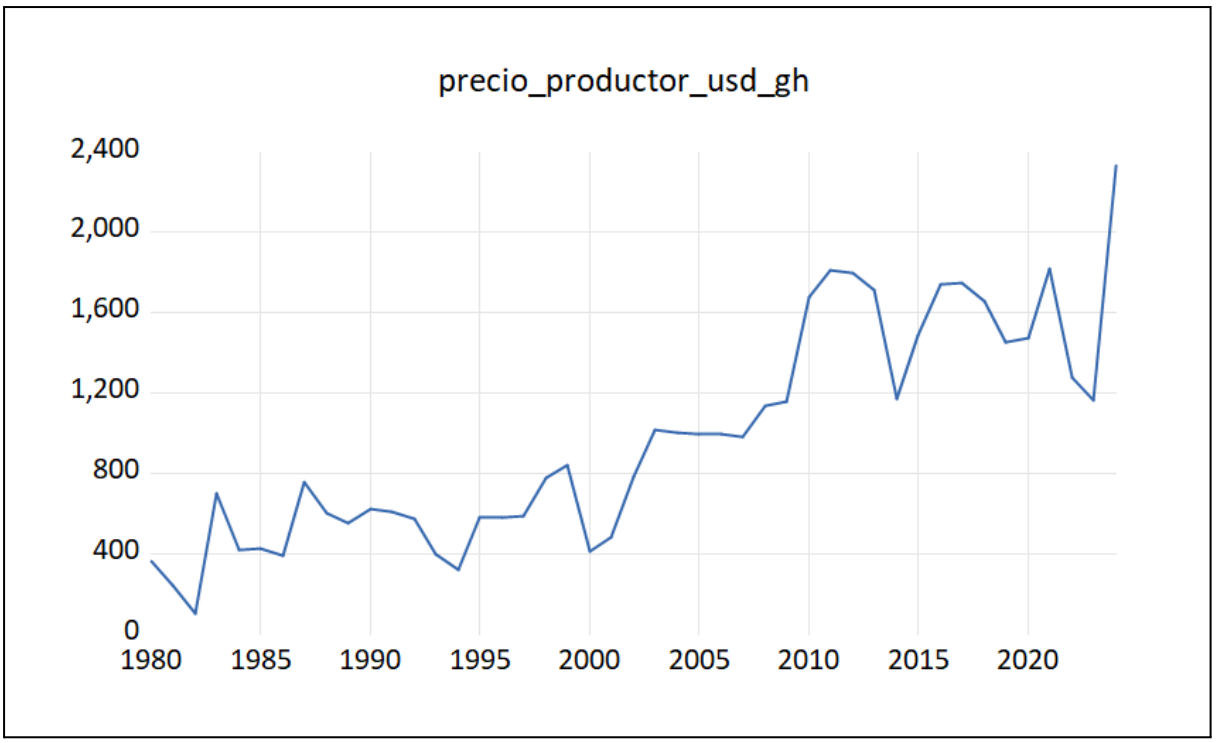


En el gráfico se observa en un principio valores estables en el precio del productor, luego una fuerte subida entre los años 2000-2010, luego cierta estabilidad, y finalmente un gran salto en 2023-2024 que puede deberse a la escasez mundial en la oferta de cacao.

Analizo los test de Dickey Fuller para verificar la existencia de raíz unitaria, empezando con el modelo con tendencia determinística e intercepto:

Null Hypothesis: PRECIO PRODUCTOR USD GH has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.985067	0.0165
Test critical values:	1% level	-4.180911	
	5% level	-3.515523	
	10% level	-3.188259	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO PRODUCTOR USD GH)
Method: Least Squares
Date: 10/14/26 Time: 18:53
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO PRODUCTOR USD GH(-1)	-0.612121	0.153604	-3.985067	0.0003
C	94.54949	81.62358	1.158360	0.2534
@TREND("1980")	23.38542	6.061770	3.857853	0.0004

R-squared	0.287137	Mean dependent var	44.89464
Adjusted R-squared	0.252363	S.D. dependent var	291.8291
S.E. of regression	252.3330	Akaike info criterion	13.96512
Sum squared resid	2610549.	Schwarz criterion	14.08677
Log likelihood	-304.2327	Hannan-Quinn criter.	14.01024
F-statistic	8.257269	Durbin-Watson stat	1.682933
Prob(F-statistic)	0.000970		

La tendencia determinista resulta significativa, por lo que observo el p-valor asociado al estadístico tau, como 0.0165<0.05 rechazo la hipótesis nula de raíz unitario, por lo que no debo diferenciar la serie.

Date: 10/14/26 Time: 18:53
Sample: 1980 2024
Included observations: 45



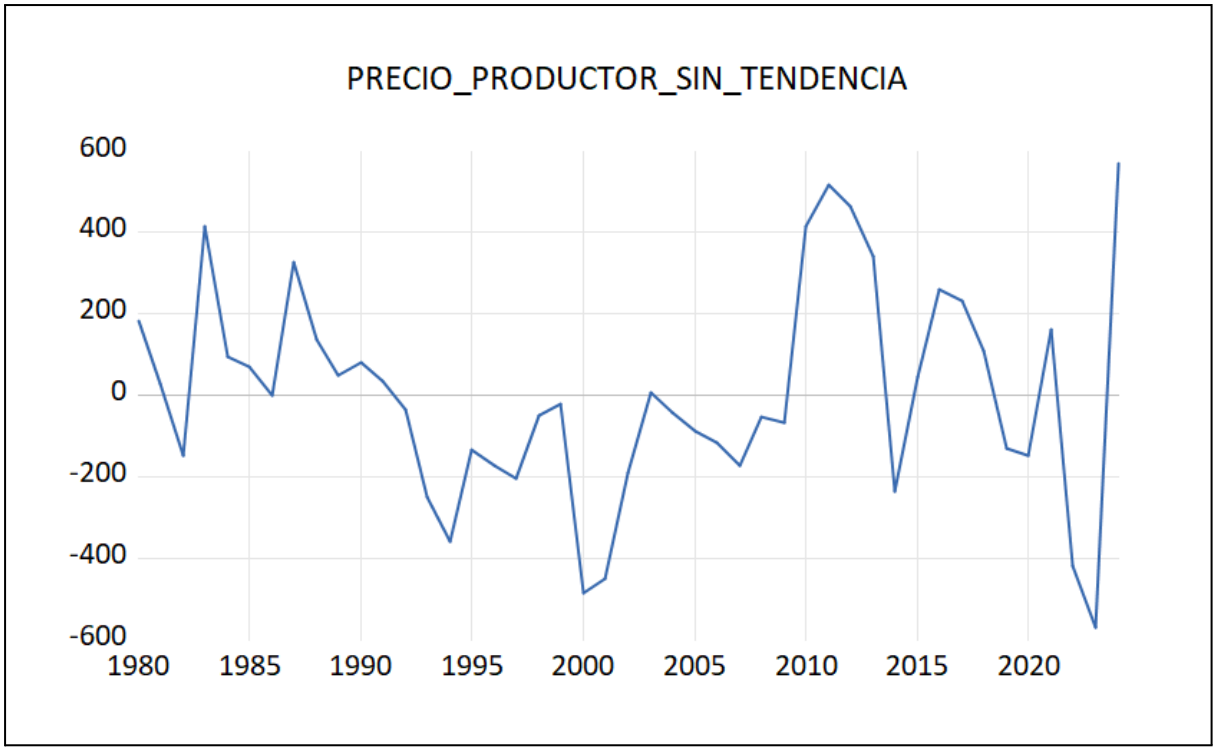
Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.769	0.769	28.423	0.000
		2	0.692	0.246	51.956	0.000
		3	0.684	0.245	75.521	0.000
		4	0.635	0.054	96.332	0.000
		5	0.596	0.043	115.10	0.000
		6	0.559	0.005	132.06	0.000
		7	0.511	-0.033	146.58	0.000
		8	0.472	-0.019	159.28	0.000
		9	0.368	-0.206	167.24	0.000
		10	0.260	-0.222	171.31	0.000
		11	0.262	0.075	175.60	0.000
		12	0.224	0.014	178.81	0.000
		13	0.175	0.030	180.84	0.000
		14	0.103	-0.105	181.57	0.000
		15	-0.014	-0.243	181.58	0.000
		16	-0.073	-0.092	181.98	0.000
		17	-0.139	-0.088	183.43	0.000
		18	-0.153	0.130	185.27	0.000
		19	-0.171	0.033	187.65	0.000
		20	-0.190	0.046	190.71	0.000

Dependent Variable: PRECIO PRODUCTOR USD GH
Method: Least Squares
Date: 10/14/26 Time: 19:00
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
@TREND	36.11294	3.058563	11.80716	0.0000
C	177.2162	78.13815	2.267985	0.0284

R-squared	0.764266	Mean dependent var	971.7009
Adjusted R-squared	0.758784	S.D. dependent var	542.5442
S.E. of regression	266.4638	Akaike info criterion	14.05178
Sum squared resid	3053128.	Schwarz criterion	14.13208
Log likelihood	-314.1651	Hannan-Quinn criter.	14.08171
F-statistic	139.4090	Durbin-Watson stat	1.200557
Prob(F-statistic)	0.000000		

Como tanto la tendencia como el intercepto resultan significativos al 5%, obtengo la serie de los residuos para analizar la serie sin tendencia determinística:



Date: 10/14/26 Time: 19:01
Sample: 1980 2024
Included observations: 45

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.341	0.341	5.5955	0.018
		2	0.072	-0.051	5.8488	0.054
		3	0.179	0.193	7.4603	0.059
		4	0.110	-0.015	8.0876	0.088
		5	0.080	0.061	8.4239	0.134
		6	0.070	-0.000	8.6864	0.192
		7	0.086	0.060	9.0988	0.246
		8	0.086	0.026	9.5238	0.300
		9	-0.132	-0.213	10.552	0.308
		10	-0.409	-0.384	20.645	0.024
		11	-0.302	-0.146	26.311	0.006
		12	-0.209	-0.124	29.117	0.004
		13	-0.111	0.114	29.929	0.005
		14	-0.035	0.112	30.013	0.008
		15	-0.150	-0.056	31.589	0.007
		16	-0.205	-0.098	34.652	0.004
		17	-0.279	-0.204	40.540	0.001
		18	-0.167	0.064	42.716	0.001
		19	-0.103	-0.086	43.583	0.001
		20	-0.054	-0.107	43.830	0.002

Se ve como cambia el correlograma con respecto al anterior, tanto la FAC como la FACP dejan de ser significativas en el segundo rezago

Dependent Variable: PRECIO_PRODUCTOR_SIN_TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/20/26 Time: 18:11
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments
Failure to improve likelihood (non-zero gradients) after 19 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1980

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	-0.463842	0.167211	-2.773991	0.0082
MA(1)	0.921565	0.060336	15.27381	0.0000

R-squared	0.230940	Mean dependent var	-4.154996
Adjusted R-squared	0.212630	S.D. dependent var	264.9680
S.E. of regression	235.1164	Akaike info criterion	13.80243
Sum squared resid	2321749.	Schwarz criterion	13.88353
Log likelihood	-301.6534	Hannan-Quinn criter.	13.83250
Durbin-Watson stat	1.566188		

Inverted AR Roots	-.46
Inverted MA Roots	-.92

Date: 10/20/26 Time: 18:12
Sample (adjusted): 1981 2024
Q-statistic probabilities adjusted for 2 ARMA terms

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.137	0.137	0.8855	
		2	0.072	0.055	1.1380	
		3	0.051	0.035	1.2646	0.261
		4	0.137	0.125	2.2154	0.330
		5	0.001	-0.038	2.2155	0.529
		6	0.079	0.071	2.5490	0.636
		7	-0.000	-0.027	2.5490	0.769
		8	0.115	0.100	3.2938	0.771
		9	-0.122	-0.156	4.1596	0.761
		10	-0.316	-0.330	10.098	0.258
		11	-0.276	-0.228	14.764	0.098
		12	-0.081	-0.050	15.179	0.126
		13	-0.094	-0.004	15.761	0.150
		14	0.093	0.237	16.350	0.176
		15	-0.163	-0.103	18.203	0.150
		16	-0.084	-0.037	18.719	0.176
		17	-0.256	-0.251	23.613	0.072
		18	-0.107	-0.061	24.496	0.079
		19	-0.086	-0.021	25.097	0.093
		20	0.023	-0.096	25.140	0.121

Ambos coeficientes son significativos y el residuo es ruido blanco, sin embargo la raíz del coeficiente MA(1) está muy cerca del círculo unitario.

Dependent Variable: PRECIO PRODUCTOR SIN TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/14/26 Time: 19:02
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments
Convergence achieved after 10 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.381693	0.149479	2.553495	0.0143

R-squared	0.131451	Mean dependent var	-4.154996
Adjusted R-squared	0.131451	S.D. dependent var	264.9680
S.E. of regression	246.9395	Akaike info criterion	13.87863
Sum squared resid	2622101.	Schwarz criterion	13.91918
Log likelihood	-304.3298	Hannan-Quinn criter.	13.89367
Durbin-Watson stat	1.669582		

Inverted AR Roots	.38

Date: 10/14/26 Time: 19:06
Sample (adjusted): 1981 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.047	0.047	0.1048	
		2	-0.158	-0.161	1.3103	0.252
		3	0.085	0.104	1.6686	0.434
		4	0.055	0.018	1.8201	0.611
		5	0.016	0.043	1.8333	0.766
		6	0.007	0.006	1.8359	0.871
		7	0.024	0.027	1.8686	0.931
		8	0.146	0.144	3.0640	0.879
		9	-0.012	-0.029	3.0729	0.930
		10	-0.345	-0.321	10.167	0.337
		11	-0.158	-0.181	11.705	0.305
		12	-0.078	-0.209	12.094	0.357
		13	-0.000	0.008	12.094	0.438
		14	0.117	0.169	13.021	0.446
		15	-0.056	0.019	13.240	0.508
		16	-0.078	-0.008	13.677	0.550
		17	-0.196	-0.259	16.566	0.414
		18	-0.065	-0.008	16.893	0.462
		19	-0.046	-0.076	17.067	0.518
		20	-0.012	-0.054	17.079	0.585

Nuevamente el coeficiente AR(1) es significativo, con raíz dentro del círculo unitario y con residuo ruido blanco, sin embargo debo probar como se comporta el coeficiente MA(1) sin el componente AR(1):

Dependent Variable: PRECIO PRODUCTOR SIN TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/14/26 Time: 19:04
Sample: 1980 2024
Included observations: 45
Failure to improve likelihood (non-zero gradients) after 15 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1979

Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.566374	0.136159	4.159653	0.0001

R-squared	0.181616	Mean dependent var	-1.09E-13
Adjusted R-squared	0.181616	S.D. dependent var	263.4184
S.E. of regression	238.3004	Akaike info criterion	13.80691
Sum squared resid	2498631.	Schwarz criterion	13.84706
Log likelihood	-309.6555	Hannan-Quinn criter.	13.82188
Durbin-Watson stat	1.841086		

Inverted MA Roots	-.57

Date: 10/14/26 Time: 19:05
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	-0.029	-0.029	0.0398	
		2	-0.029	-0.030	0.0810	0.776
		3	0.091	0.089	0.4953	0.781
		4	0.078	0.083	0.8117	0.847
		5	0.011	0.022	0.8183	0.936
		6	0.025	0.023	0.8526	0.974
		7	0.044	0.032	0.9592	0.987
		8	0.088	0.084	1.4002	0.986
		9	0.008	0.010	1.4036	0.994
		10	-0.310	-0.323	7.2047	0.616
		11	-0.148	-0.222	8.5642	0.574
		12	-0.058	-0.135	8.7793	0.642
		13	-0.062	-0.042	9.0304	0.700
		14	0.109	0.214	9.8366	0.707
		15	-0.104	-0.015	10.600	0.717
		16	-0.055	-0.012	10.823	0.765
		17	-0.215	-0.256	14.314	0.575
		18	-0.049	-0.049	14.506	0.631
		19	-0.071	-0.021	14.914	0.668
		20	-0.035	-0.076	15.018	0.721

El coeficiente MA(1) por separado arroja el problema de “Failure to improve likelihood…”, lo cual puede deberse a la poca cantidad de observaciones, por lo que este modelo podría ser problemático.

Sin embargo si se pasa por alto eso y se quiere comparar ambos modelos, necesito comparar con el criterio de información Akaike, el cual requiere que ambos modelos tengan la misma cantidad de observaciones, por lo que estimo nuevamente el MA(1) con 44 observaciones:

Dependent Variable: PRECIO PRODUCTOR SIN TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/14/26 Time: 19:06
Sample: 1981 2024
Included observations: 44
Failure to improve likelihood (non-zero gradients) after 22 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1980

Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.603754	0.122461	4.930154	0.0000

R-squared	0.179885	Mean dependent var	-4.154996
Adjusted R-squared	0.179885	S.D. dependent var	264.9680
S.E. of regression	239.9555	Akaike info criterion	13.82125
Sum squared resid	2475882.	Schwarz criterion	13.86180
Log likelihood	-303.0675	Hannan-Quinn criter.	13.83629
Durbin-Watson stat	1.903025		

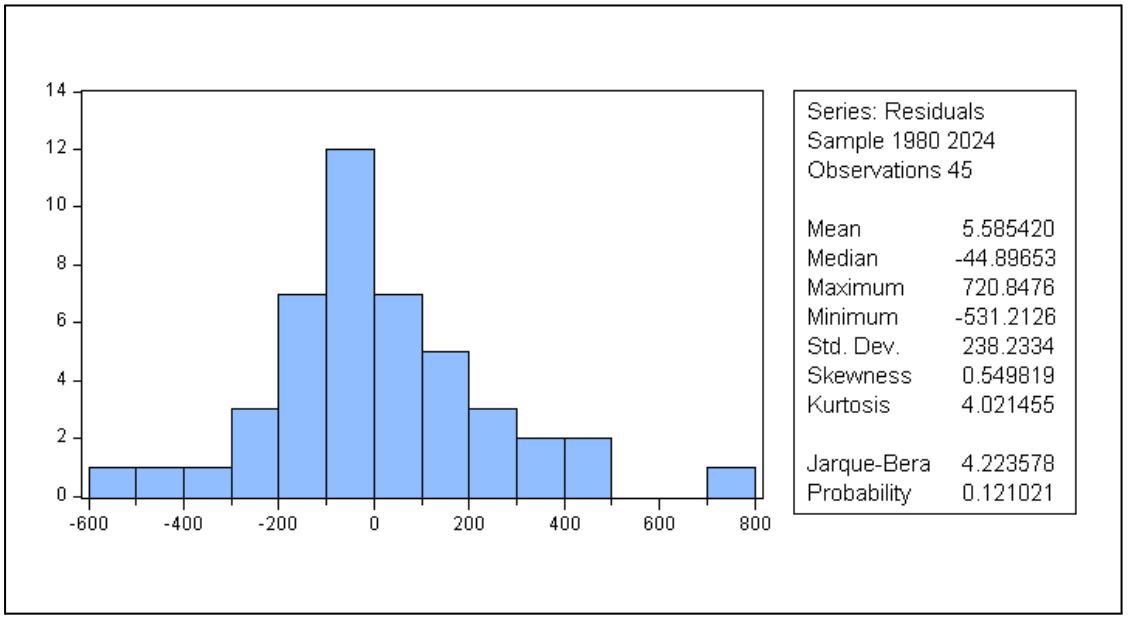
Inverted MA Roots	-.60

Date: 10/14/26 Time: 19:07
Sample: 1981 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	-0.056	-0.056	0.1453	
		2	-0.005	-0.008	0.1466	0.702
		3	0.042	0.041	0.2321	0.890
		4	0.099	0.105	0.7325	0.866
		5	-0.010	0.002	0.7378	0.947
		6	0.035	0.035	0.8042	0.977
		7	0.014	0.010	0.8155	0.992
		8	0.091	0.085	1.2857	0.989
		9	0.004	0.013	1.2868	0.996
		10	-0.304	-0.318	6.8032	0.658
		11	-0.147	-0.222	8.1347	0.616
		12	-0.046	-0.116	8.2683	0.689
		13	-0.054	-0.052	8.4560	0.749
		14	0.132	0.236	9.6353	0.723
		15	-0.111	-0.022	10.497	0.725
		16	-0.034	-0.004	10.580	0.782
		17	-0.211	-0.252	13.921	0.605
		18	-0.043	-0.084	14.065	0.663
		19	-0.072	-0.020	14.486	0.697
		20	-0.002	-0.080	14.486	0.755

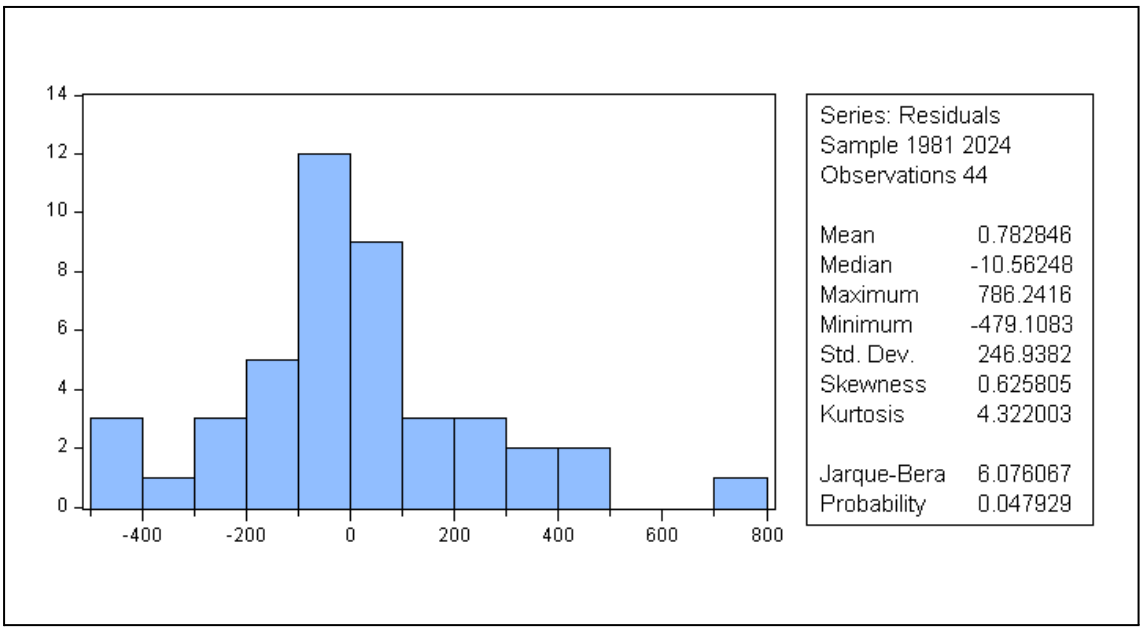
La nueva estimación del MA(1) sigue arrojando el problema de convergencia, sin embargo tiene un mejor AIC que el AR(1), 13.82<13.87.
Por lo que si ignoramos el problema de la falla de convergencia, el modelo elegido sería el MA(1), de lo contrario el modelo sería el AR(1)

Jarque-Bera test MA(1)



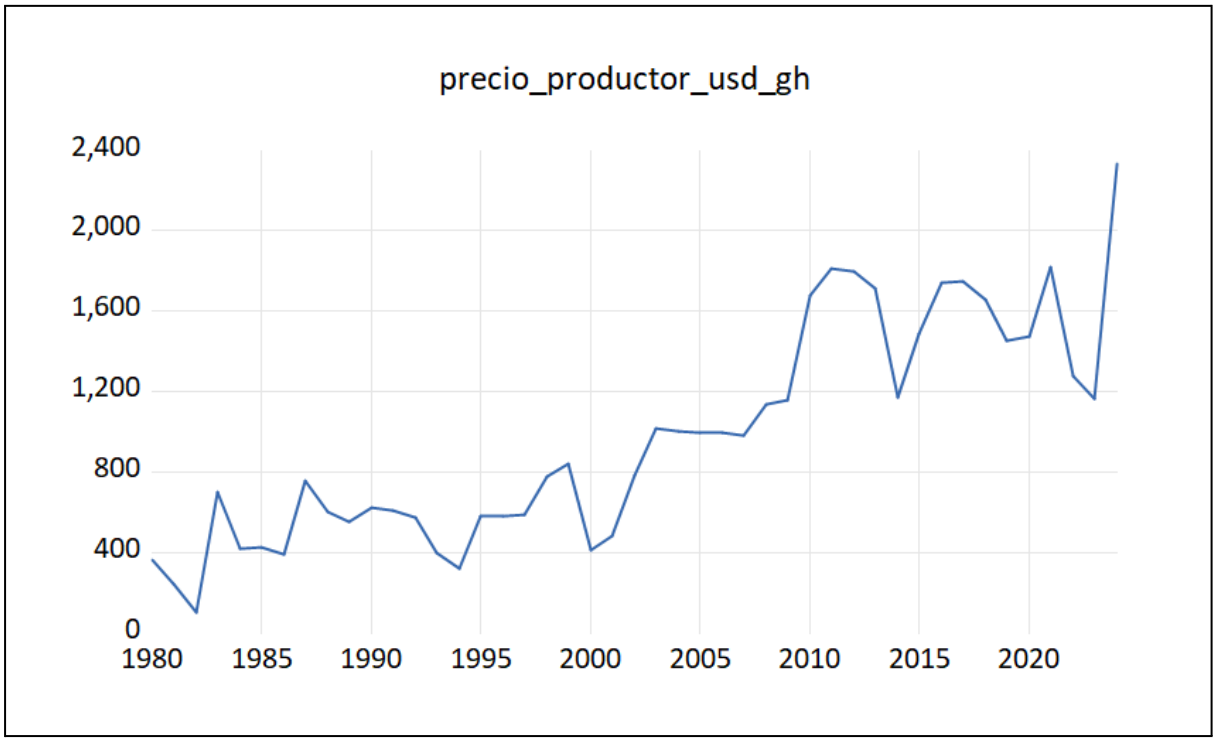
No se rechaza la hipótesis nula, por lo que los residuos del MA(1) se comportan normalmente.

Jarque-Bera test AR(1)



Se rechaza por muy poco (0.047929<0.05) el test de normalidad de residuos de Jarque-Bera con un nivel de significancia del 5%.

Agregando una Variable de Impacto:



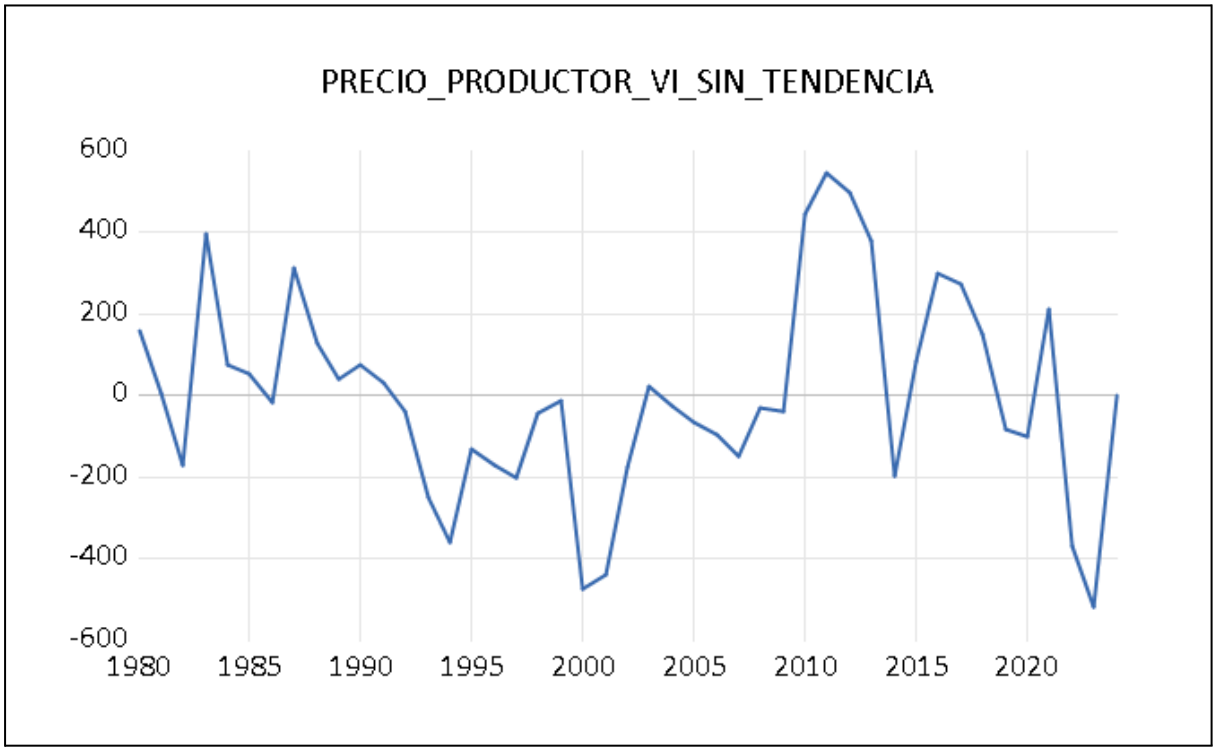
Durante los últimos años está ocurriendo una crisis respecto a la producción del cacao en África Occidental, provocada por un virus sobre la planta de cacao, que afectó a Ghana y Costa de Marfil, dos de los mayores productores de cacao a nivel mundial (60% de la oferta mundial).
Es por eso que decidí incorporar una variable de intervencion que refleje ese shock tanto en la producción como en el precio internacional, y por lo tanto en el precio del productor.
Como se puede ver en el primer gráfico de la serie, esto se vio reflejado en la serie con un gran salto durante el anio 2024, debido a que el gobierno Ghanes debió reajustar el precio del productor a mitad de anio debido a la gran diferencia que había con el precio internacional.

Dependent Variable: PRECIO_PRODUCTOR_USD_GH
Method: Least Squares
Date: 11/02/26 Time: 13:05
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	203.0896	75.14324	2.702699	0.0099
@TREND	34.30782	3.009343	11.40044	0.0000
VI_2024	622.7663	265.1383	2.348836	0.0236

R-squared	0.791636	Mean dependent var	971.7009
Adjusted R-squared	0.781714	S.D. dependent var	542.5442
S.E. of regression	253.4824	Akaike info criterion	13.97281
Sum squared resid	2698641.	Schwarz criterion	14.09325
Log likelihood	-311.3881	Hannan-Quinn criter.	14.01771
F-statistic	79.78529	Durbin-Watson stat	0.976569
Prob(F-statistic)	0.000000		

La variable de intervención (toma valor 0 en todos los años y 1 en 2024) resultó significativa junto a la tendencia y el intercepto



Null Hypothesis: PRECIO_PRODUCTOR_VI_SIN_TENDENCIA has a unit root
Exogenous: None
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.773091	0.0004
Test critical values:	1% level	-2.618579	
	5% level	-1.948495	
	10% level	-1.612135	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO_PRODUCTOR_VI_SIN_TENDENCIA)
Method: Least Squares
Date: 11/02/26 Time: 13:07
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO_PRODUCTOR_VI_SIN_TENDENC	-0.492848	0.130622	-3.773091	0.0005

R-squared	0.248568	Mean dependent var	-3.566965
Adjusted R-squared	0.248568	S.D. dependent var	247.5390
S.E. of regression	214.5797	Akaike info criterion	13.59770
Sum squared resid	1979911.	Schwarz criterion	13.63825
Log likelihood	-298.1495	Hannan-Quinn criter.	13.61274
Durbin-Watson stat	1.841784		

Luego de quitar la tendencia e intercepto a la serie, y quitar el impacto del año 2024 a la serie, verificamos que el test de dickey fuller rechaza la presencia de raíz unitaria.

Date: 11/03/26 Time: 09:50
Sample: 1980 2024
Included observations: 45

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.507	0.507	12.363	0.000
		2	0.179	-0.105	13.937	0.001
		3	0.173	0.172	15.452	0.001
		4	0.168	0.023	16.916	0.002
		5	0.134	0.043	17.859	0.003
		6	0.071	-0.031	18.133	0.006
		7	0.059	0.032	18.327	0.011
		8	0.052	-0.010	18.481	0.018
		9	-0.149	-0.255	19.780	0.019
		10	-0.397	-0.319	29.306	0.001
		11	-0.403	-0.149	39.417	0.000
		12	-0.332	-0.146	46.500	0.000
		13	-0.242	0.013	50.383	0.000
		14	-0.147	0.097	51.849	0.000
		15	-0.177	-0.032	54.064	0.000
		16	-0.244	-0.080	58.390	0.000
		17	-0.301	-0.142	65.251	0.000
		18	-0.186	0.093	67.973	0.000
		19	-0.121	-0.123	69.160	0.000
		20	-0.074	-0.083	69.627	0.000

Dependent Variable: PRECIO_PRODUCTOR_VI_SIN_TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 11/02/26 Time: 13:11
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments
Convergence achieved after 11 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.507152	0.130622	3.882592	0.0004

R-squared	0.259417	Mean dependent var	-3.566965
Adjusted R-squared	0.259417	S.D. dependent var	249.3456
S.E. of regression	214.5797	Akaike info criterion	13.59770
Sum squared resid	1979911.	Schwarz criterion	13.63825
Log likelihood	-298.1495	Hannan-Quinn criter.	13.61274
Durbin-Watson stat	1.841784		

Inverted AR Roots .51

Date: 11/02/26 Time: 13:13
Sample (adjusted): 1981 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.060	0.060	0.1675	
		2	-0.150	-0.154	1.2525	0.263
		3	0.019	0.040	1.2709	0.530
		4	0.086	0.061	1.6475	0.649
		5	0.062	0.063	1.8485	0.764
		6	-0.008	0.005	1.8516	0.869
		7	-0.009	0.005	1.8563	0.932
		8	0.153	0.149	3.1707	0.869
		9	-0.015	-0.047	3.1837	0.922
		10	-0.304	-0.276	8.6978	0.466
		11	-0.186	-0.190	10.830	0.371
		12	-0.120	-0.236	11.738	0.384
		13	-0.066	-0.146	12.026	0.444
		14	0.057	0.060	12.242	0.508
		15	-0.040	0.020	12.352	0.578
		16	-0.080	0.004	12.813	0.617
		17	-0.212	-0.197	16.170	0.441
		18	-0.034	0.042	16.259	0.506
		19	-0.029	-0.070	16.325	0.570
		20	-0.010	-0.055	16.333	0.635

Dependent Variable: PRECIO_PRODUCTOR_VI_SIN_TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/02/26 Time: 13:12
Sample: 1980 2024
Included observations: 45
Convergence achieved after 11 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.536318	0.132053	4.061395	0.0002
SIGMASQ	43937.87	9089.990	4.833654	0.0000

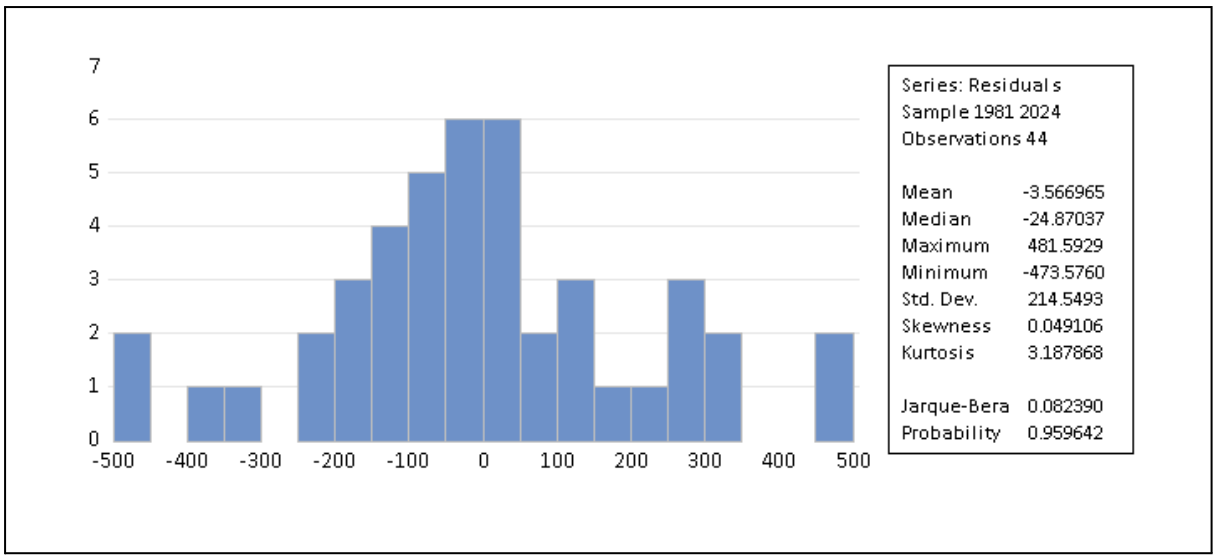
R-squared	0.267333	Mean dependent var	2.69E-13
Adjusted R-squared	0.250294	S.D. dependent var	247.6545
S.E. of regression	214.4330	Akaike info criterion	13.62483
Sum squared resid	1977204.	Schwarz criterion	13.70513
Log likelihood	-304.5588	Hannan-Quinn criter.	13.65477
Durbin-Watson stat	1.838321		

Inverted MA Roots -.54

Date: 11/02/26 Time: 13:12
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.071	0.071	0.2454	
		2	0.119	0.115	0.9477	0.330
		3	0.068	0.053	1.1792	0.555
		4	0.127	0.108	2.0114	0.570
		5	0.069	0.043	2.2608	0.688
		6	0.039	0.005	2.3444	0.800
		7	0.015	-0.012	2.3573	0.884
		8	0.080	0.059	2.7269	0.909
		9	-0.059	-0.084	2.9325	0.939
		10	-0.295	-0.324	8.2034	0.514
		11	-0.236	-0.243	11.673	0.308
		12	-0.177	-0.156	13.671	0.252
		13	-0.160	-0.119	15.373	0.222
		14	-0.018	0.110	15.395	0.283
		15	-0.124	0.027	16.480	0.285
		16	-0.101	-0.001	17.231	0.305
		17	-0.252	-0.195	22.021	0.143
		18	-0.057	0.018	22.277	0.174
		19	-0.093	-0.023	22.976	0.192
		20	-0.037	-0.095	23.092	0.233

Comparando los modelos, ambos resultan significativos, con ruido blanco y raíces dentro del círculo unitario, por lo que los comparo con el criterio de información de Akaike. Según el cual elijo el modelo **AR(1)** (13.59<13.62)



Con el nuevo modelo incorporando la variable de intervención pude obtener un modelo con menor varianza, menor criterio de Akaike y con residuos ruido blanco, en comparación con el mejor modelo sin variable de intervención.